

Project Name - terraform-aws-vpc-ec2-with-variables

COMPLETE STEP-BY-STEP TERRAFORM DOCUMENTATION (VPC + EC2)

Project Goal

Create AWS Infrastructure using Terraform:

- VPC
- Public Subnet
- Private Subnet
- Internet Gateway
- Route Tables
- Security Group
- EC2 Instance (Free Tier)
- Variables (string, number, bool, list, tuple, map, object, any)
- Dynamic Tags

Region: **us-west-1**

STEP 1: Launch EC2 (Terraform Server)

1. Go to AWS Console
2. Open EC2 → Launch Instance
3. Select:
 - AMI: Amazon Linux 2
 - Instance Type: t3.micro (Free Tier)
4. Create Key Pair
5. Allow SSH (Port 22)
6. Launch Instance
7. Connect via SSH:

```
ssh -i key.pem ec2-user@your-public-ip
```

STEP 2: Install Terraform on EC2

Run these commands:

```
sudo yum update -y
sudo yum install -y yum-utils
sudo yum-config-manager --add-repo
https://rpm.releases.hashicorp.com/AmazonLinux/hashicorp.repo
sudo yum -y install terraform
```

Check version:

```
terraform -v
```

STEP 3: Create Project Folder

```
mkdir terraform-project
cd terraform-project
```

STEP 4: Create Required Files (Best Practice Structure)

Create files:

```
touch provider.tf
touch variables.tf
touch vpc.tf
touch ec2.tf
```

Project structure:

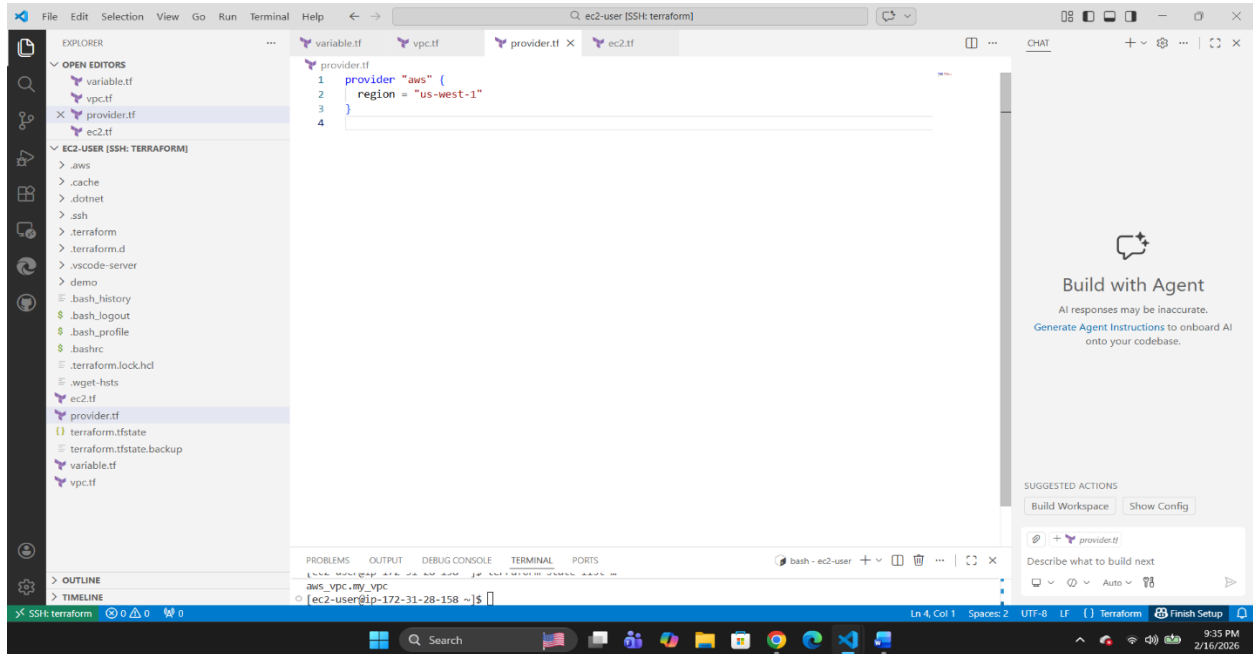
```
terraform-project/
├── provider.tf
├── variables.tf
├── vpc.tf
└── ec2.tf
```

STEP 5: Provider Configuration (provider.tf)

```
provider "aws" {

  region = "us-west-1"
```

}



STEP 6: Variables File (variables.tf)

Includes all types: string, number, bool, list, tuple, map, object, any + tags

```
variable "aws_region" {  
  type    = string  
  default = "us-west-1"  
}
```

```
variable "vpc_name" {  
  type    = string  
  default = "vpc-1"  
}
```

```
variable "ssh_port" {  
  type    = number  
  default = 22  
}
```

```
variable "associate_public_ip" {  
  type    = bool
```

```

    default = true
}

variable "private_subnet_cidr" {
    type    = list(string)
    default = ["10.0.4.0/23"]
}

variable "instance_config" {
    type    = tuple([string, string])
    default = ["us-west-1a", "t3.micro"]
}

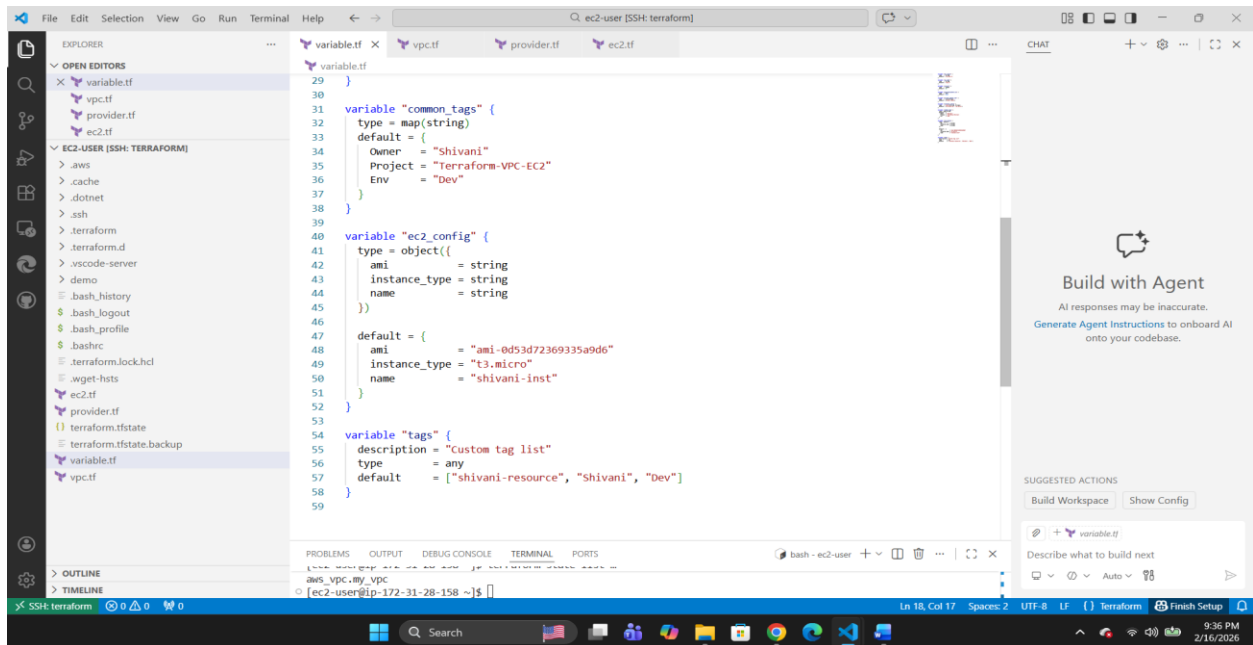
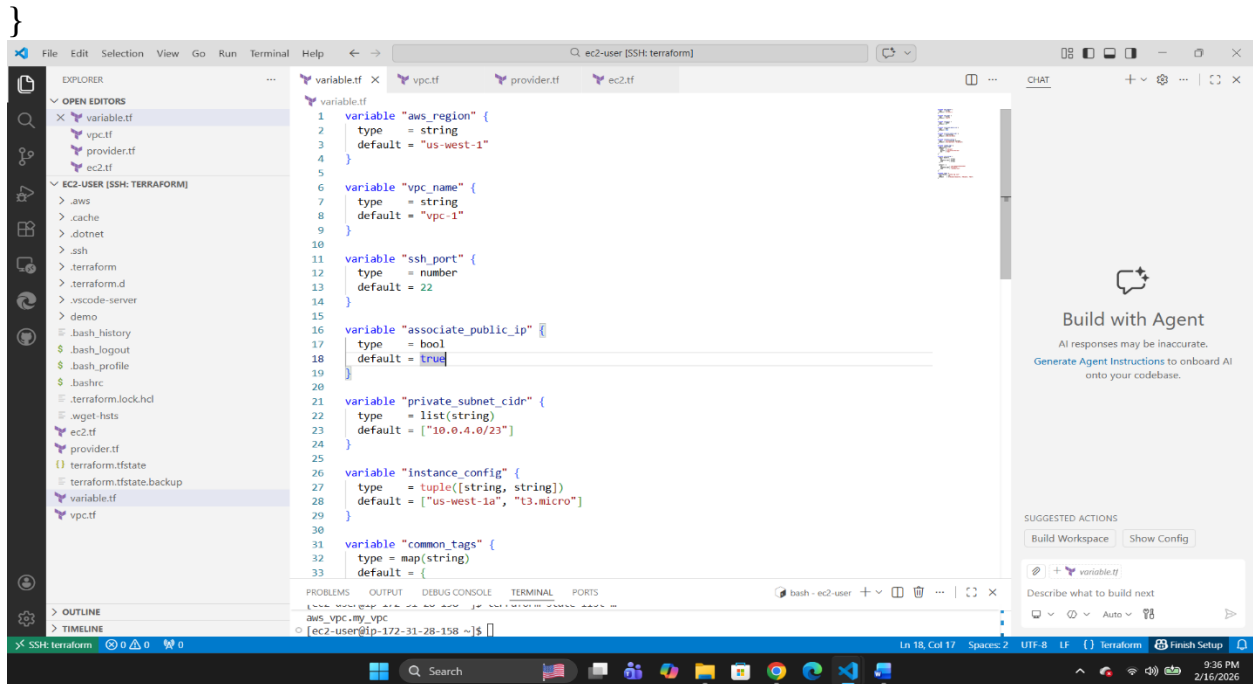
variable "common_tags" {
    type = map(string)
    default = {
        Owner   = "Shivani"
        Project = "Terraform-VPC-EC2"
        Env     = "Dev"
    }
}

variable "ec2_config" {
    type = object({
        ami            = string
        instance_type = string
        name           = string
    })

    default = {
        ami            = "ami-0d53d72369335a9d6"
        instance_type = "t3.micro"
        name           = "shivani-inst"
    }
}

variable "tags" {
    description = "Custom tag list"
    type        = any
    default     = ["shivani-resource", "Shivani", "Dev"]
}

```



STEP 7: VPC Infrastructure (vpc.tf)

```
resource "aws_vpc" "my_vpc" {
  cidr_block = "10.0.0.0/21"
```

```
tags = {
```

```

    Name = var.vpc_name
    Env  = var.tags[2]
    Owner = var.tags[1]
  }
}

resource "aws_subnet" "pub_sub" {
  vpc_id      = aws_vpc.my_vpc.id
  cidr_block  = "10.0.0.0/22"
  availability_zone = var.instance_config[0]

  tags = {
    Name = "${var.tags[0]}-public"
    Env  = var.tags[2]
    Owner = var.tags[1]
  }
}

resource "aws_subnet" "pri_sub" {
  vpc_id      = aws_vpc.my_vpc.id
  cidr_block  = var.private_subnet_cidr[0]
  availability_zone = "us-west-1c"

  tags = {
    Name = "${var.tags[0]}-private"
    Env  = var.tags[2]
    Owner = var.tags[1]
  }
}

resource "aws_internet_gateway" "my_igw" {
  vpc_id = aws_vpc.my_vpc.id

  tags = {
    Name = "${var.tags[0]}-igw"
    Env  = var.tags[2]
    Owner = var.tags[1]
  }
}

```

```

resource "aws_route_table" "my_rt" {
  vpc_id = aws_vpc.my_vpc.id

  tags = {
    Name = "${var.tags[0]}-private-rt"
    Env  = var.tags[2]
    Owner = var.tags[1]
  }
}

resource "aws_route_table_association" "my_prta" {
  route_table_id = aws_route_table.my_rt.id
  subnet_id      = aws_subnet.pri_sub.id
}

resource "aws_route_table" "my_pubrt" {
  vpc_id = aws_vpc.my_vpc.id

  tags = {
    Name = "${var.tags[0]}-public-rt"
    Env  = var.tags[2]
    Owner = var.tags[1]
  }
}

resource "aws_route" "public_route" {
  route_table_id      = aws_route_table.my_pubrt.id
  destination_cidr_block = "0.0.0.0/0"
  gateway_id          = aws_internet_gateway.my_igw.id
}

resource "aws_route_table_association" "my_pubrta" {
  route_table_id = aws_route_table.my_pubrt.id
  subnet_id      = aws_subnet.pub_sub.id
}

resource "aws_security_group" "my_sg" {
  name       = "my_sg_1"
  description = "allow ssh access"
  vpc_id     = aws_vpc.my_vpc.id
}

```

```

ingress {
  from_port = var.ssh_port
  to_port   = var.ssh_port
  protocol  = "tcp"
  cidr_blocks = ["0.0.0.0/0"]
}

```

```

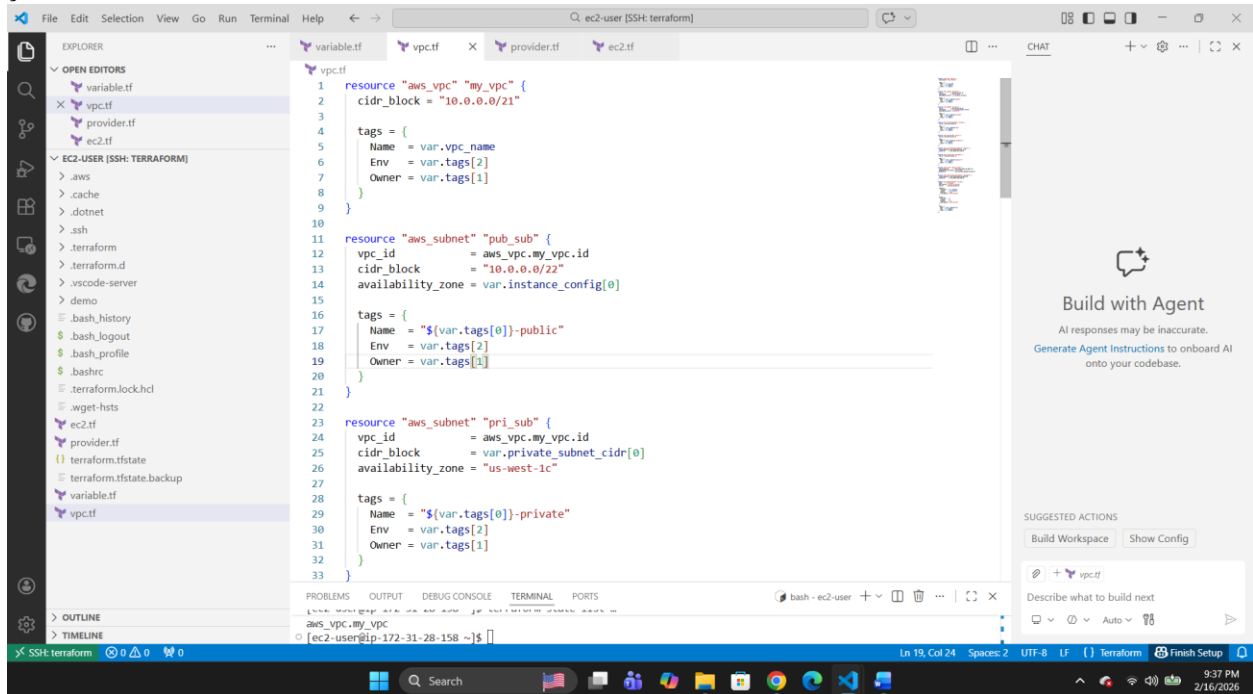
egress {
  from_port = 0
  to_port   = 0
  protocol  = "-1"
  cidr_blocks = ["0.0.0.0/0"]
}

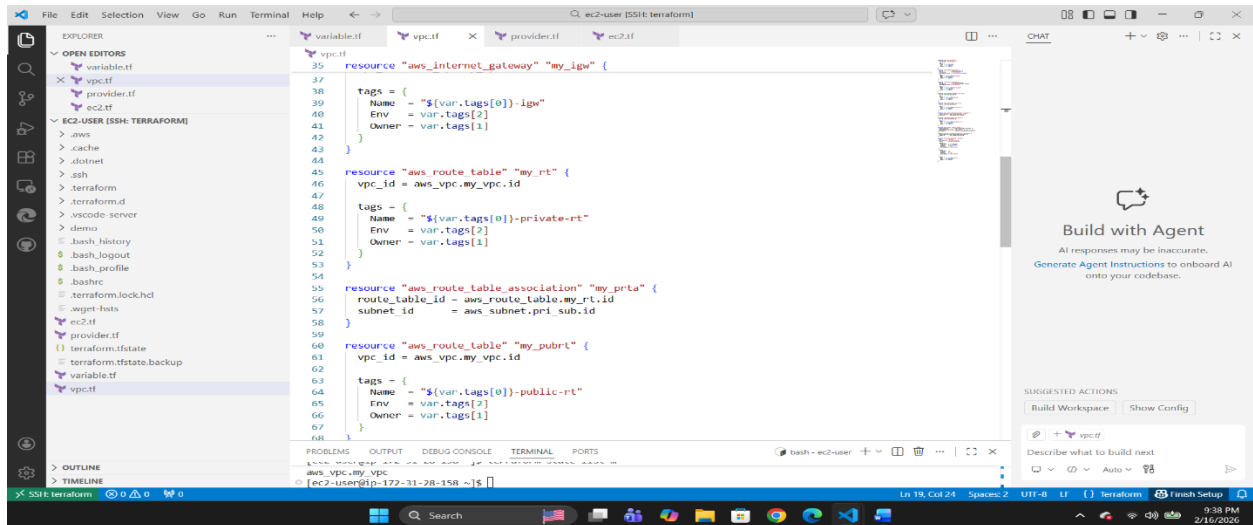
```

```

tags = {
  Name = "${var.tags[0]}-sg"
  Env  = var.tags[2]
  Owner = var.tags[1]
}

```





STEP 8: EC2 Instance (ec2.tf)

```

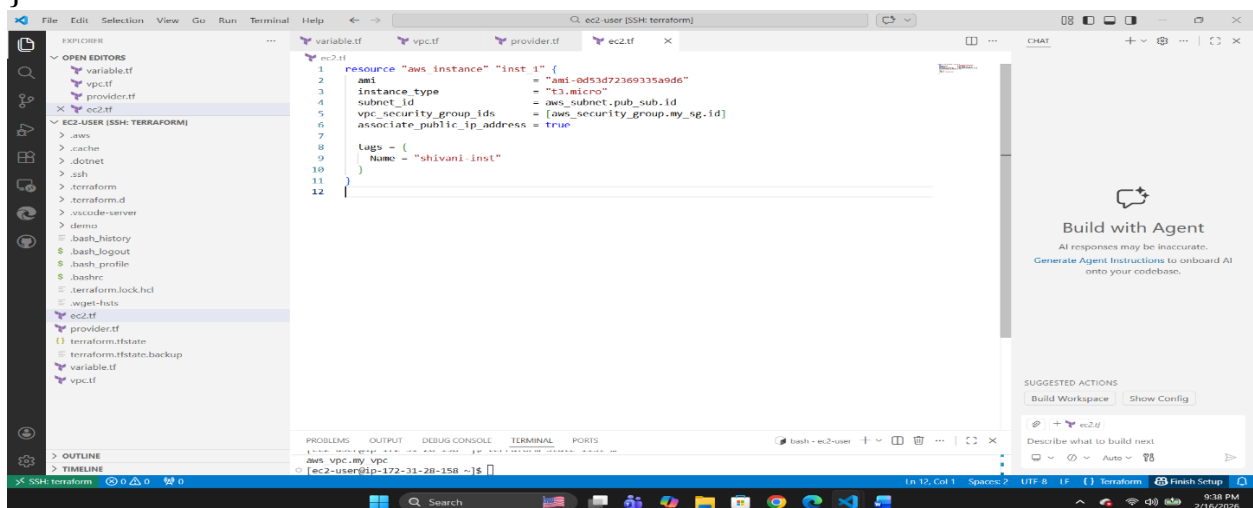
resource "aws_instance" "inst_1" {
  ami           = "ami-0d53d72369335a9d6"
  instance_type = "t3.micro"
  subnet_id     = aws_subnet.pub_sub.id
  vpc_security_group_ids = [aws_security_group.my_sg.id]
  associate_public_ip_address = true

```

```

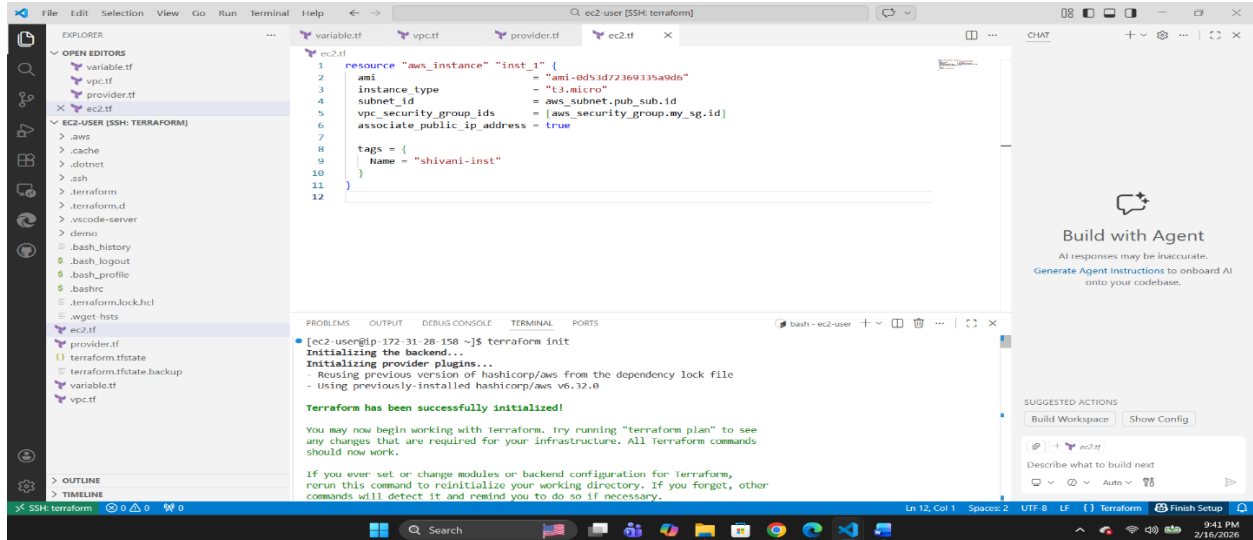
  tags = {
    Name = "shivani-inst"
  }
}

```



STEP 9: Initialize Terraform

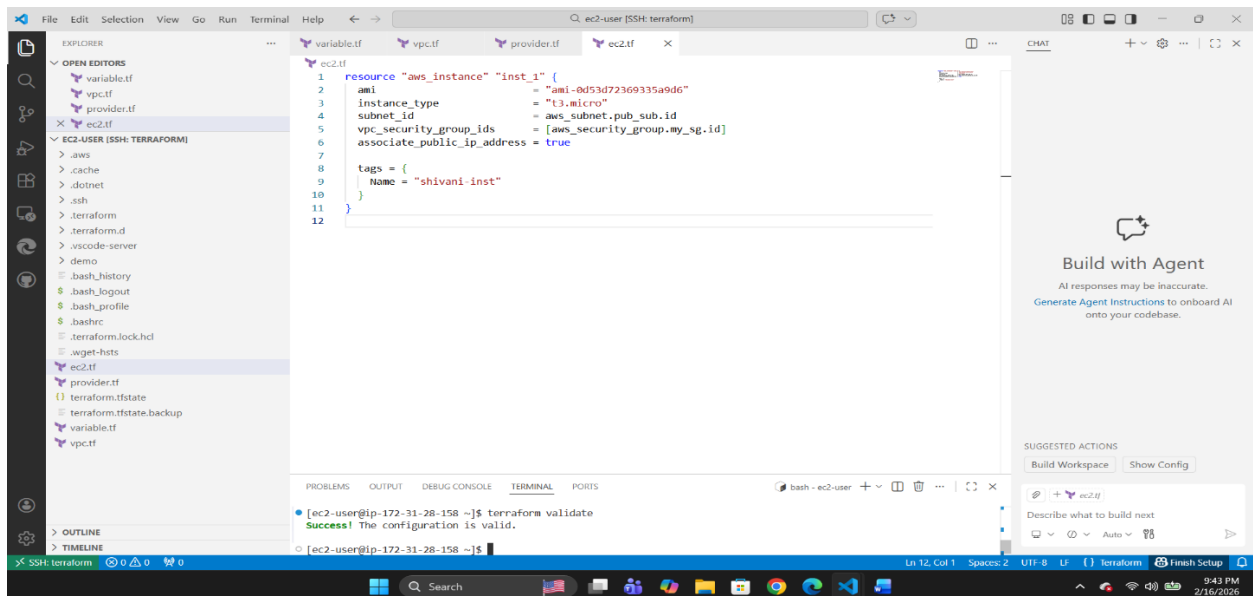
terraform init



This downloads AWS provider plugins.

STEP 10: Validate Configuration

terraform validate



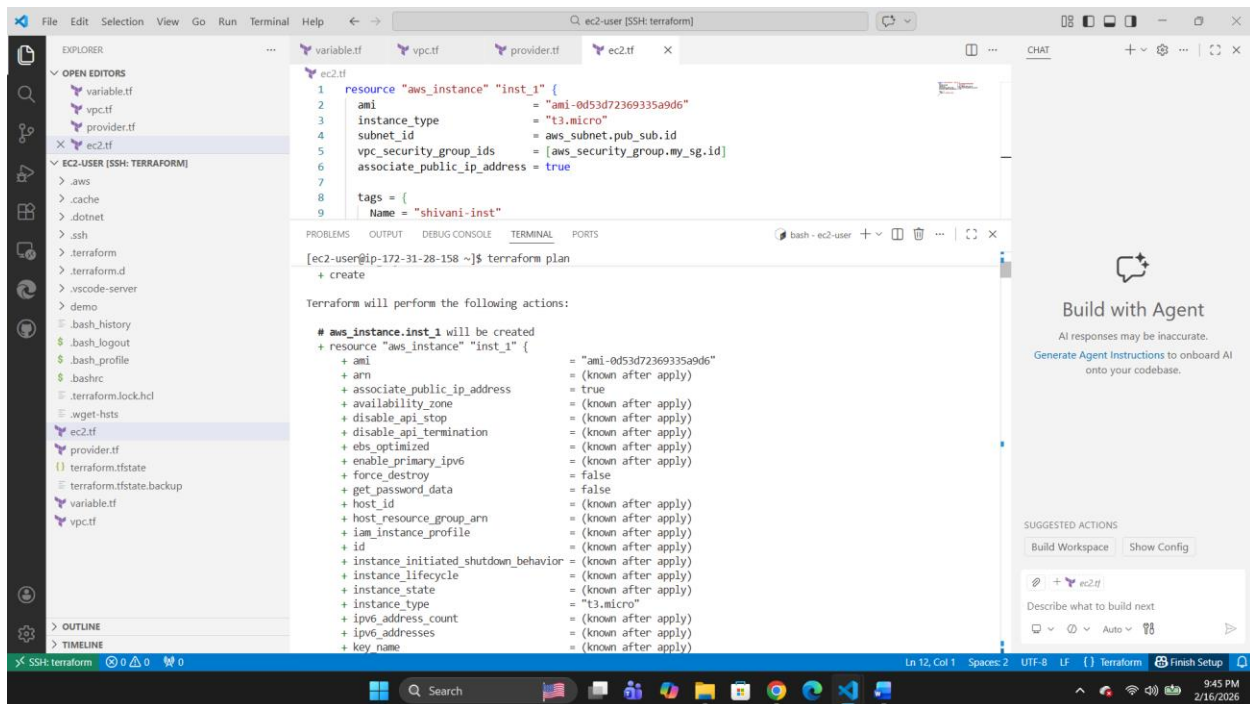
Checks syntax errors (bool, duplicate resource, etc.)

STEP 11: Preview Infrastructure

```
terraform plan
```

Shows what resources will be created:

- VPC
- Subnets
- IGW
- Route Tables
- Security Group
- EC2

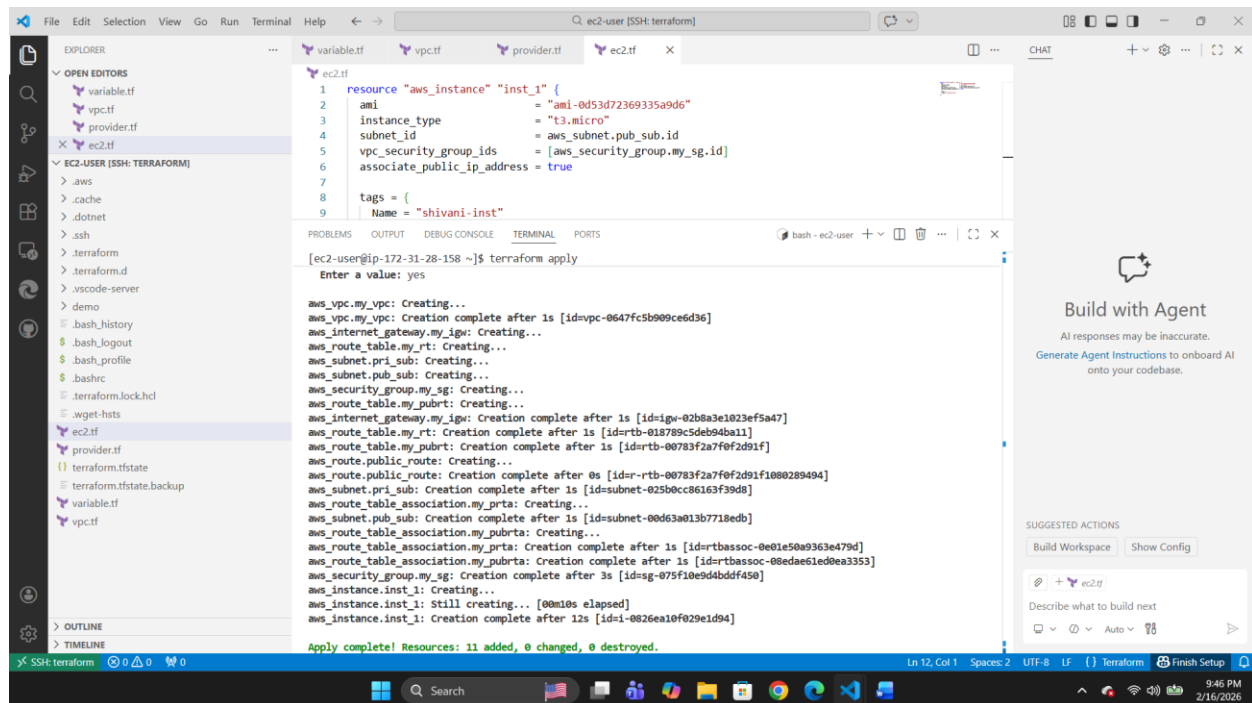


STEP 12: Deploy Infrastructure

```
terraform apply
```

Type:

Yes



The screenshot shows a VS Code editor with a terminal window open. The terminal is running the command `terraform apply` in an `ec2-user` shell. The output shows the creation of various AWS resources, including VPC, subnets, security groups, route tables, and an EC2 instance. The resources are created successfully, and the terminal shows the final state of the infrastructure.

```
variable.tf  vpc.tf  provider.tf  ec2.tf  x
EXPLORER
  OPEN EDITORS
    variable.tf
    vpc.tf
    provider.tf
    ec2.tf
  EC2-USER [SSH: TERRAFORM]
    .aws
    .cache
    .dotnet
    .ssh
    terraform
    terraform.d
    vscode-server
    demo
    .bash_history
    .bash_logout
    .bash_profile
    .bashrc
    terraform.lock.hcl
    .wget-hsts
    ec2.tf
    provider.tf
    terraform.tfstate
    terraform.tfstate.backup
    variable.tf
    vpc.tf
  OUTLINE
  TIMELINE
[ec2-user@ip-172-31-28-158 ~]$ terraform apply
Enter a value: yes

aws_vpc.my_vpc: Creating...
aws_vpc.my_vpc: Creation complete after 1s [id=vpc-0647fc5b909ce6d36]
aws_internet_gateway.my_igw: Creating...
aws_route_table.my_rt: Creating...
aws_subnet.pri_sub: Creating...
aws_subnet.pub_sub: Creating...
aws_security_group.my_sg: Creating...
aws_route_table.my_pubrt: Creating...
aws_internet_gateway.my_igw: Creation complete after 1s [id=igw-02b8a3e1023ef5a47]
aws_route_table.my_rt: Creation complete after 1s [id=rtb-018789c5de94ba11]
aws_route_table.my_pubrt: Creation complete after 1s [id=rtb-00783f2a7f0f2d91f]
aws_route.public_route: Creating...
aws_route.public_route: Creation complete after 0s [id=rtb-00783f2a7f0f2d91f080289494]
aws_subnet.pri_sub: Creation complete after 1s [id=subnet-025b0cc86163f39d8]
aws_route_table_association.my_prta: Creating...
aws_subnet.pub_sub: Creation complete after 1s [id=subnet-00d63a013b7718edb]
aws_route_table_association.my_prta: Creation complete after 1s [id=rtbassoc-0e01e50a9363e479d]
aws_route_table_association.my_pubrta: Creation complete after 3s [id=rtbassoc-0e01e50a9363e479d]
aws_security_group.my_sg: Creation complete after 3s [id=sg-075f10e9d4bddf450]
aws_instance.inst_1: Creating...
aws_instance.inst_1: Still creating... [00m10s elapsed]
aws_instance.inst_1: Creation complete after 12s [id=i-0826ea10f029e1d94]

Apply complete! Resources: 11 added, 0 changed, 0 destroyed.
```

Now AWS resources will be created.

STEP 13: Verify Resources

Check state:

```
terraform state list
```

Expected output:

```
aws_instance.inst_1
```

```
aws_internet_gateway.my_igw
```

```
aws_route.public_route
```

```
aws_route_table.my_pubrt
```

```
aws_route_table.my_rt
```

```
aws_route_table_association.my_prta
```

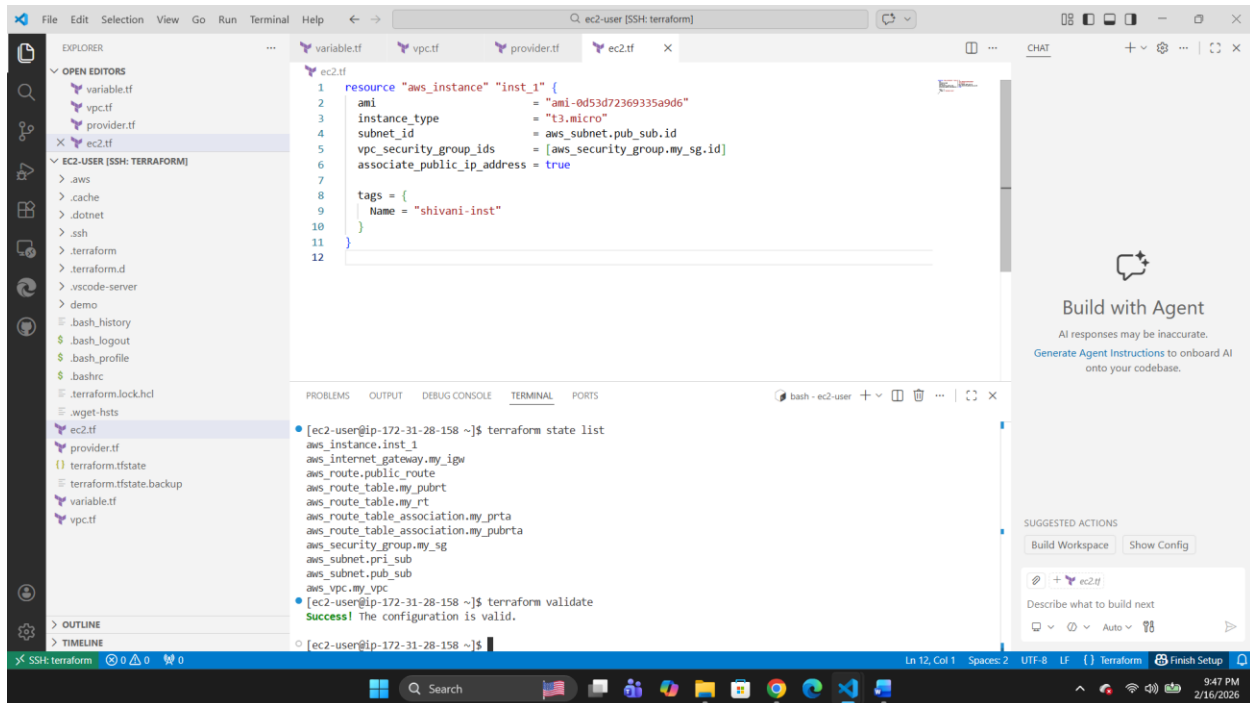
aws_route_table_association.my_pubrta

aws_security_group.my_sg

aws_subnet.pri_sub

aws_subnet.pub_sub

aws_vpc.my_vpc



STEP 14: Check in AWS Console

Go to:

1. AWS Console → VPC → Your VPC

us-west-1.console.aws.amazon.com/vpcconsole/home?region=us-west-1#vpcs

varsha berya (0850-1319-2054)

varsha berya

VPC > Your VPCs

VPC dashboard <

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Your VPCs

VPCs VPC encryption controls

Your VPCs (1/4) Info

Find VPCs by attribute or tag

Last updated 31 minutes ago

Actions Create VPC

Name	VPC ID	State	Encryption c...	Encryption control ...	Block Public...	IPv...
vpc-1	vpc-0647fc5b909ce6d36	Available	-	-	Off	10.0.0.0/22

vpc-0647fc5b909ce6d36 / vpc-1

Details Resource map CIDRs Flow logs Tags Integrations

Tags

Search tags

Key	Value
Owner	Shivani
Env	Dev
Name	vpc-1

Manage tags

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2. Subnets section

us-west-1.console.aws.amazon.com/vpcconsole/home?region=us-west-1#subnets

varsha berya (0850-1319-2054)

varsha berya

VPC > Subnets

VPC dashboard <

AWS Global View

Filter by VPC

Virtual private cloud

Your VPCs

Subnets

Route tables

Internet gateways

Egress-only internet gateways

DHCP option sets

Elastic IPs

Managed prefix lists

NAT gateways

Peering connections

Route servers

Security

Network ACLs

Security groups

Subnets (1/7) Info

Find subnets by attribute or tag

Last updated 31 minutes ago

Actions Create subnet

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
kops-private-subnet	subnet-0923d127c588c3ac9	Available	vpc-0064dba44a715c874 kop...	Off	10.0.2.0/24
shivani-resource-private	subnet-025b0cc86163f39d8	Available	vpc-0647fc5b909ce6d36 vpc-1	Off	10.0.4.0/23
kops-public-subnet	subnet-06d24f54fd57cde5d	Available	vpc-0064dba44a715c874 kop...	Off	10.0.1.0/24
shivani-resource-public	subnet-00d63a013b7718edb	Available	vpc-0647fc5b909ce6d36 vpc-1	Off	10.0.0.0/22

subnet-025b0cc86163f39d8 / shivani-resource-private

Details Flow logs Route table Network ACL CIDR reservations Sharing Tags

Tags (3)

Search tags

Key	Value
Owner	Shivani
Env	Dev
Name	shivani-resource-private

Manage tags

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Subnets (1/7) Info

Find subnets by attribute or tag

Name	Subnet ID	State	VPC	Block Public...	IPv4 CIDR
☐ kops-private-subnet	subnet-0923d127c588c3ac9	Available	vpc-0064dba44a715c874 kop...	Off	10.0.2.0/24
☐ shivani-resource-private	subnet-025b0cc86163f39d8	Available	vpc-0647fc5b909ce6d36 vpc-1	Off	10.0.4.0/23
☐ kops-public-subnet	subnet-06d24f54fd57cde5d	Available	vpc-0064dba44a715c874 kop...	Off	10.0.1.0/24
☒ shivani-resource-public	subnet-00d63a013b7718edb	Available	vpc-0647fc5b909ce6d36 vpc-1	Off	10.0.0.0/22

subnet-00d63a013b7718edb / shivani-resource-public

Details | Flow logs | Route table | Network ACL | CIDR reservations | Sharing | Tags

Tags (3)

Key	Value
Owner	Shivani
Env	Dev
Name	shivani-resource-public

3. Route tables - shivani-resource-private-rt

Route tables (1/8) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
☒ shivani-resource-private-rt	rtb-018789c5deb94ba11	subnet-025b0cc86163f3...	-	No	vpc-0647fc5b909ce6d36 vpc
☐ -	rtb-0393ce2047f7c90ae	-	-	Yes	vpc-0ad3859899fe7dc32
☐ shivani.k8s.local	rtb-011d45615ae6d7898	subnet-0ae90c64c32785...	-	No	vpc-03ba095d83250b10d sh
☐ -	rtb-08a54c18bf7113a1d	-	-	Yes	vpc-0647fc5b909ce6d36 vpc

rtb-018789c5deb94ba11 / shivani-resource-private-rt

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Tags (3)

Key	Value
Owner	Shivani
Env	Dev
Name	shivani-resource-private-rt

shivani-resource-public-rt

Route tables (1/8) Info

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC
shivani-resource-public-rt	rtb-07e4cab80df2dc80e	subnet-06d24f54fd57cde...	-	No	vpc-0064dba44a715c874 kops-vpc-us...
kops-public-rt	rtb-03e32b400f82ba130	-	-	Yes	vpc-0064dba44a715c874 kops-vpc-us...
-	rtb-044511ac9153fadb4	-	-	Yes	vpc-03ba095d83250b10d shivani.k8s.local

rtb-00783f2a7f0f2d91f / shivani-resource-public-rt

Details | Routes | Subnet associations | Edge associations | Route propagation | **Tags**

Tags (3)

Search tags

Key	Value
Name	shivani-resource-public-rt
Env	Dev
Owner	Shivani

Manage tags

4. Internet gateways - shivani-resource-igw

Internet gateways (1/4) Info

Find internet gateways by attribute or tag

Name	Internet gateway ID	State	VPC ID	Owner
shivani-resource-igw	igw-02b8a3e1023ef5a47	Attached	vpc-0647fc5b909ce6d36 vpc-1	085013192054
kops-igw	igw-0437bc4f50122eefe	Attached	vpc-0064dba44a715c874 kops-vpc-us...	085013192054
shivani.k8s.local	igw-0b8e7c5ed115ca7df	Attached	vpc-03ba095d83250b10d shivani.k8s...	085013192054
-	igw-0d5478e36b23f8015	Attached	vpc-0ad3859899fe7dc32	085013192054

igw-02b8a3e1023ef5a47 / shivani-resource-igw

Details | **Tags**

Tags (3)

Search tags

Key	Value
Env	Dev
Name	shivani-resource-igw
Owner	Shivani

Manage tags

4. Security Groups

The screenshot shows the AWS VPC console. On the left, the 'VPC dashboard' sidebar is visible with categories like 'Virtual private cloud' and 'Security'. The main area displays 'Security Groups (1/15)'. A table lists several security groups, with 'shivani-resource-sg' (ID: sg-075f10e9d4bddf450) selected. Below the table, the details for 'sg-075f10e9d4bddf450 - my_sg_1' are shown, including a 'Tags' section with a table:

Key	Value
Env	Dev
Name	shivani-resource-sg
Owner	Shivani

5. EC2 → Instances

Region must be **us-west-1** (very important)

The screenshot shows the AWS EC2 console. On the left, the 'EC2' sidebar is visible with categories like 'Dashboard', 'Instances', 'Images', and 'Elastic Block Store'. The main area displays 'Instances (1/2)'. A table lists two instances, with 'shivani-inst' (ID: i-0826ea10f029e1d94) selected. Below the table, the details for 'i-0826ea10f029e1d94 (shivani-inst)' are shown, including a 'Tags' section with a table:

Key	Value
Name	shivani-inst